

FAKE ACCOUNT DETECTION IN SOCIAL MEDIA

Using Petri Net Simulation for Twitter/X Account Analysis

Social Network Security (SNS) – Project Report

Submitted By

Vaitheeswaran S

Roll No: 2303717710421305

Social Media Platform: Twitter/X

Detection Target: Fake / Bot Accounts

Method: Petri Net

Simulation Tool: TINA 4.0

Academic Demonstration using Synthetic Twitter/X Account Data

Table of Contents

- 1. Abstract**
- 2. Introduction**
- 3. Problem Statement**
- 4. Objectives**
- 5. Dataset and Social Media Platform**
- 6. Characteristics Analysis**
- 7. Feature Selection**
- 8. Proposed System**
- 9. Petri Net Model**
- 10. TINA 4.0 Simulation**
- 11. Evaluation and Performance Analysis**
- 12. Advantages**
- 13. Limitations**
- 14. Future Enhancement**
- 15. Conclusion**
- 16. References**

1. Abstract

Social media platforms such as Twitter/X contain genuine users, automated bots, spam accounts, and potentially fraudulent accounts. Identifying suspicious accounts is important for social-network security. This project proposes a Petri Net-based approach for modelling fake-account detection. Account age, follower count, following count, posting activity, and profile completeness are analyzed. The workflow is represented using a Petri Net and simulated using TINA 4.0. Synthetic Twitter/X account data is used for demonstration and evaluation.

2. Introduction

Social media platforms are widely used for communication and information sharing. Because of their scale, they can also contain fake, automated, spam, or malicious accounts. Automated account analysis can help organize the detection process by examining account metadata and behavioural characteristics.

This project models the detection workflow as a Petri Net. A token represents an account moving through analysis states. TINA 4.0 is used to simulate profile analysis, behaviour analysis, bot detection, and final classification.

3. Problem Statement

The increasing number of fake and automated accounts on social media platforms makes it difficult to distinguish genuine users from suspicious accounts. Therefore, there is a need for an efficient and systematic method to analyze account characteristics and behaviour and identify potentially fake accounts.

4. Objectives

Main Objective: To develop a Petri Net-based model for detecting potentially fake Twitter/X accounts based on account and behavioural characteristics.

Specific Objectives:

5. Dataset and Social Media Platform

Social Media Platform: Twitter/X

The project focuses on Twitter/X because it has been widely studied for bot and fake-account detection. Public research datasets such as TwiBot-20 and TwiBot-22 contain account, tweet, and relationship information useful for research.

Dataset used in this academic demonstration: A synthetic Twitter/X account dataset is used. It is not claimed to be a live collection of real Twitter/X users. Its feature structure is based on commonly used account-level characteristics.

Important: This project demonstrates the detection workflow and Petri Net simulation rather than claiming real-world Twitter/X detection performance.

Feature	Description
Account ID	Unique account identifier
Account Age	Age in days
Followers	Number of followers

Feature	Description
Following	Number of accounts followed
Posts	Number of posts
Profile Complete	Whether profile information is complete
Activity	Approximate activity level
Expected Class	Fake/Bot or Genuine/Human

Synthetic demonstration rule: (Account Age < 30 days) AND (Following > 1000) AND (Profile Complete = No) → Fake. Otherwise → Genuine/Real. This is only an academic rule, not a complete real-world detector.

6. Characteristics Analysis

Account Age: Very new accounts may require additional scrutiny when combined with other signals.

Followers and Following: An unusual relationship between these counts can provide an account-level signal.

Posting Activity: Extremely high, repetitive, or unusual activity can be associated with automated behaviour.

Profile Completeness: Missing or incomplete profile information can be one of several signals.

Behaviour: Repeated actions, excessive activity, and automated interaction patterns can provide additional signals.

7. Feature Selection

The selected features are Account Age, Followers, Following, Posts, Profile Completeness, and Activity.

These features represent account history, network characteristics, profile information, and activity behaviour.

8. Proposed System

The proposed system receives Twitter/X account data, analyzes selected characteristics, passes the account through a Petri Net workflow, and produces a final classification.

System Flow: Twitter/X Account Data → Characteristics Analysis → Feature Selection → Petri Net Model → Profile Analysis → Behaviour Analysis → Bot Detection → Classification → Genuine/Human or Fake/Bot

9. Petri Net Model

A Petri Net represents states, events, and token flow. In this project it represents the stages of fake-account analysis.

Place / State: Represents a state such as Account_Created or Behaviour_Analyzed.

Transition: Represents an operation such as Profile_Analysis or Bot_Detection_Check.

Token: Represents the account currently being processed.

Arc: Connects places and transitions and represents process flow.

Implemented workflow: Account_Created → Profile_Analysis → Profile_Analyzed → Behaviour_Analysis → Behaviour_Analyzed → Bot_Detection_Check → Bot_Detection → Classify_Fake → Fake_Account

The final place in this simple deterministic workflow demonstrates a classification path. A production model would normally include separate final branches for Fake/Bot and Genuine/Human.

10. TINA 4.0 Simulation

TINA 4.0 is used to create and simulate the Petri Net. The graphical editor allows places, transitions, arcs, and token markings to be observed.

At the beginning, one token is placed in Account_Created. As transitions fire, the token moves through profile analysis, behaviour analysis, bot detection, and classification.

The following screenshot shows the implemented Petri Net model in TINA 4.0. The model contains Account_Created, Profile_Analysis, Profile_Analyzed, Behaviour_Analysis, Behaviour_Analyzed, Bot_Detection_Check, Bot_Detection, Classify_Fake, and Fake_Account states/transitions.

TINA's stepper simulator is step-based. A firing sequence can be observed one step at a time or represented through a saved scenario/history for replay.

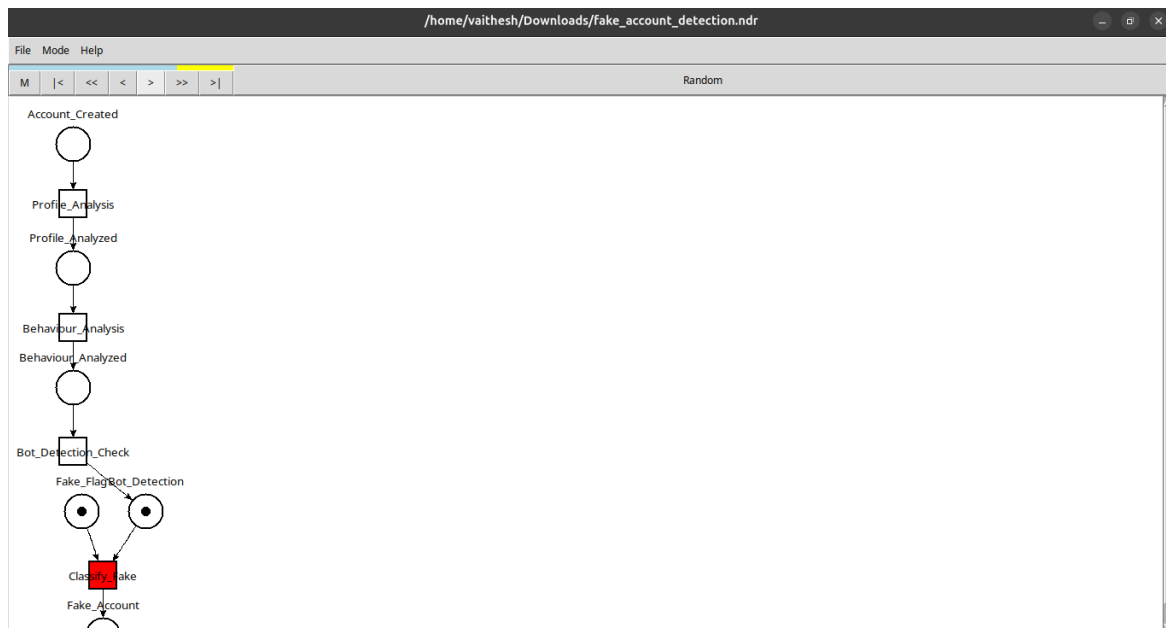


Figure 1. TINA 4.0 graphical Petri Net simulation model.

11. Evaluation and Performance Analysis

Relevant metrics include Accuracy, Precision, Recall, F1-score, False Positive Rate, False Negative Rate, and processing/transition time.

For the synthetic demonstration, 100% accuracy can occur when labels are constructed using exactly the same predefined rules used by the simulator. This demonstrates workflow correctness only and must not be presented as real-world Twitter/X detection performance.

For real-world evaluation, an independently labelled dataset and unseen test data should be used.

12. Advantages

- Simple visual representation.
- Clear sequential processing using states and transitions.
- Account token movement can be observed.
- TINA provides practical Petri Net simulation.
- Model can be extended with additional stages.
- Can support integration of advanced detection methods.

13. Limitations

- Current demonstration uses synthetic data.
- It does not directly access live Twitter/X data.
- Limited number of account features.
- Real-world detection is more complex than the simplified rule.
- A suspicious feature does not prove that an account is fake.
- Real systems may require ML, graph analysis, NLP, and continuously updated data.
- Synthetic 100% accuracy must not be generalized to real populations.

14. Future Enhancement

- Integrate a suitable public Twitter/X bot-detection dataset.
- Add machine-learning classification.
- Add more behavioural and network features.
- Add graph-based relationship analysis.
- Add NLP-based post analysis.
- Develop real-time monitoring.
- Extend the Petri Net with separate Genuine/Human and Fake/Bot branches.
- Compare the rule-based Petri Net with models such as Random Forest.

15. Conclusion

This project demonstrates the use of a Petri Net for modelling fake-account detection on Twitter/X. The system analyzes account characteristics and behaviour through states and transitions, while TINA 4.0 provides simulation and visualization. The model can be extended with real-world datasets, machine learning, NLP, and graph-based analysis.

The present implementation is an academic simulation based on synthetic Twitter/X account data, so its evaluation demonstrates the predefined workflow rather than production-level detection effectiveness.

16. References

TINA – Time Petri Net Analyzer, LAAS-CNRS. TINA 4.0 documentation and manuals.

Twibot-20 and Twibot-22 research literature on Twitter bot detection.

Research literature on malicious/fake profile detection and Petri-net-based modelling for social-network security.

Twitter/X public documentation, where applicable.